Momentum Analysis

Empirical Analysis of Momentum and Reversal Effects

Dataset: `Dataset_PS-A.csv`

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1 Objective

The objective of this analysis is to investigate whether the given stock-price dataset exhibits a **momentum effect**.

In a conventional momentum strategy, stocks that have performed well in the recent past are expected to continue performing well in the near future, while stocks that have performed poorly are expected to continue performing poorly.

Thus, the central research question is:

Does past stock performance predict future stock performance?

More specifically, we investigate whether returns over different historical lookback periods have a positive relationship with future returns over different forward horizons.

A positive relationship would support conventional momentum, whereas a negative relationship would be evidence consistent with **short- or medium-term reversal**.

2 Dataset and Initial Preparation

The analysis was performed using:

`Dataset_PS-A.csv`

The dataset contains daily OHLCV-type stock information for multiple stocks identified by their ticker symbols.

The analysis covered:

- 24 tickers,
- daily observations,
- historical price data from which daily returns and momentum signals were calculated.

2.1 Date Conversion

The date column was converted into a proper datetime format.

This is necessary because momentum analysis is inherently time-dependent. Observations must be correctly ordered before historical and future returns can be calculated.

For example, the required chronological ordering is:

$$P_{t-L} \longrightarrow P_t \longrightarrow P_{t+H},$$

where L is the lookback period and H is the forward horizon.

Inference

The data was converted into a suitable chronological format, allowing time-series calculations to be performed correctly.

2.2 Sorting by Stock and Date

The dataset was sorted by

$$(\text{Ticker}, \text{Date}).$$

This ensures that observations belonging to each stock appear in chronological order independently of the other stocks.

Inference

This ensures that returns and momentum signals are calculated within each individual stock and that observations from different stocks are not mixed.

3 Calculation of Daily Returns

Daily close-to-close returns were calculated using

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1,$$

where

- $P_{i,t}$ is the closing price of stock i at time t,
- $R_{i,t}$ is the daily return of stock i.

The first observation for every stock has no previous price and therefore produces a missing return.

Several unusually large daily movements were present in the dataset. Examples include approximately

$$-27.91\%, \qquad +27.69\%, \qquad -17.85\%, \qquad -23.23\%.$$

These observations were retained rather than being arbitrarily removed.

Inference

The dataset contains substantial return variation and several extreme observations.

Such observations can increase volatility and influence statistical relationships. However, removing them without a documented methodological reason could itself introduce bias.

4 Definition of Momentum

Momentum was measured using cumulative historical returns.

For a lookback period L, momentum was defined as

$$M_{i,t}^{(L)} = \frac{P_{i,t}}{P_{i,t-L}} - 1.$$

The following lookback periods were tested:

$$L \in \{5, 10, 20, 60, 120\}.$$

Thus, for each stock and date, the following momentum measures were calculated:

- 5-day momentum,
- 10-day momentum,
- 20-day momentum,
- 60-day momentum,
- 120-day momentum.

For example, if

$$M_{i,t}^{(20)} = 0.10,$$

then stock i gained approximately 10% over the previous 20 trading days.

If conventional momentum exists, stocks with larger values of $M_{i,t}^{(L)}$ should subsequently produce larger future returns.

Inference

Testing multiple lookback periods avoids assuming that a particular definition of "recent performance" is necessarily correct.

It also allows us to determine whether any observed effect is persistent across different historical horizons.

5 Definition of Future Returns

Future returns were calculated over four different horizons:

$$H \in \{1, 5, 10, 20\}.$$

The forward return was defined as

$$F_{i,t}^{(H)} = \frac{P_{i,t+H}}{P_{i,t}} - 1.$$

Therefore, the analysis tested relationships such as

$$\text{5-day momentum} \rightarrow \text{1-day future return},$$

and similarly for the other combinations of lookback and forward horizons.

Inference

Using multiple forward horizons allows us to determine whether the observed relationship is concentrated in the very short term or persists over longer periods.

6 Initial 20-Day Momentum Test

The first major test examined the relationship

$$\text{20-day momentum} \rightarrow \text{1-day future return.}$$

The relationship was measured using the daily **cross-sectional Spearman Information Coefficient (IC)**.

For each date, the IC measures the rank correlation between

$$\text{past 20-day momentum}$$

and

$$\text{next-day return.}$$

The results were:

Table 1: Initial 20-Day Momentum Test

Statistic	**Value**
Mean IC	-0.0196485
Median IC	-0.0208696
IC Standard Deviation	0.2710384
Positive IC Fraction	46.68%
t-statistic	-2.77281
p-value	0.005628

6.1 Interpretation

The mean IC was

$$\boxed{\text{IC} = -0.01965}.$$

The negative sign indicates that, on average, stocks with higher past 20-day returns tended to have lower next-day returns.

The positive IC fraction was only

$$46.68\%,$$

meaning that fewer than half of the daily cross-sectional observations produced a positive relationship.

The test statistic was

$$t = -2.77$$

with

$$p = 0.00563.$$

Inference

The initial test does not support conventional momentum.

Instead, it provides evidence of a negative relationship between past performance and subsequent performance, which is consistent with a possible short-term reversal effect.

However, the magnitude of the IC is small, so this result alone does not establish that the effect is economically large or tradable.

7 Momentum Lookback × Forward Horizon Analysis

To determine whether the observed negative relationship was specific to the 20-day/1-day combination, the analysis was expanded.

The following lookback periods were tested:

$$5,\ 10,\ 20,\ 60,\ 120$$

and the following forward horizons:

$$1,\ 5,\ 10,\ 20.$$

This resulted in

$$5 \times 4 = 20$$

different momentum configurations.

The mean IC values were:

Table 2: Mean Information Coefficient Across Lookback and Forward Horizons

Lookback	**1 Day**	**5 Days**	**10 Days**	**20 Days**
5 days	−0.0243	−0.0234	−0.0201	−0.0305
10 days	−0.0219	−0.0218	−0.0292	−0.0437
20 days	−0.0196	−0.0247	−0.0371	**−0.0673**
60 days	−0.0158	−0.0222	−0.0346	−0.0421
120 days	−0.0035	−0.0046	−0.0097	−0.0052

7.1 Interpretation

An important observation is that all 20 mean IC values were negative.

There was therefore no tested combination in which the average cross-sectional relationship became positive.

The strongest negative relationship occurred for

$$\text{20-day lookback} \rightarrow \text{20-day forward return},$$

with

$$\boxed{\text{IC} = -0.0673}.$$

Other relatively strong negative relationships included

$$\mathrm{IC}_{10,20} = -0.0437,$$

$$\mathrm{IC}_{60,20} = -0.0421,$$

and

$$\mathrm{IC}_{20,10} = -0.0371.$$

Inference

The negative relationship is not restricted to the original 20-day/1-day test.

Across the tested horizons, higher past returns generally corresponded to lower subsequent returns.

This pattern is more consistent with reversal than with conventional momentum.

8 Statistical Significance Across Horizons

The strength of the relationship varied across different horizons.

For the

$$\text{20-day lookback} \rightarrow \text{20-day forward}$$

combination, the test statistic was approximately

$$t = -10.4124,$$

with

$$p \approx 1.57 \times 10^{-24}.$$

In contrast, the 120-day lookback produced much weaker evidence.

For example,

$$p \approx 0.637$$

for the 120-day lookback with a 1-day forward horizon, while

$$p \approx 0.509$$

was obtained for the 120-day lookback with a 20-day forward horizon.

Inference

The reversal relationship appears to be concentrated more strongly in shorter and medium-term lookback periods.

The 120-day signal does not provide convincing evidence of the same relationship.

9 Interpretation of IC Magnitudes

The direction and magnitude of the IC must be distinguished.

For example,

$$\text{IC} = -0.0196$$

is statistically detectable but economically modest.

Statistical significance does not automatically imply that the effect is sufficiently large to generate a profitable trading strategy.

Therefore, a portfolio-level analysis was performed.

Inference

The IC analysis establishes that the dataset contains a measurable negative cross-sectional relationship, but a portfolio test is required to determine whether this relationship translates into an economically meaningful return spread.

10 Quantile Portfolio Analysis

Stocks were ranked according to their past momentum and divided into five groups.

The groups can be interpreted as:

- Group 0: lowest momentum,
- Group 1: relatively low momentum,
- Group 2: middle momentum,
- Group 3: relatively high momentum,
- Group 4: highest momentum.

For the important

$$\text{20-day momentum} \rightarrow \text{20-day forward return}$$

case, the average forward returns were:

Table 3: Forward Returns by Momentum Quantile

Momentum Group	**Mean Forward Return**
Lowest momentum	2.1763%
Group 2	2.0155%
Group 3	1.7570%
Group 4	1.4788%
Highest momentum	0.9887%

10.1 Interpretation

The relationship is approximately monotonic:

$$2.1763\% > 2.0155\% > 1.7570\% > 1.4788\% > 0.9887\%.$$

As historical momentum increases, subsequent average returns decrease.

The lowest-momentum group subsequently achieved the highest average return, while the highest-momentum group achieved the lowest average return.

The difference between the lowest- and highest-momentum groups was

$$2.1763\% - 0.9887\% = \boxed{1.1876\%}.$$

Inference

The quantile portfolio analysis provides clearer economic evidence of reversal.

Rather than merely observing a small negative correlation, we observe a systematic decline in future returns as past momentum increases.

11 Long–Short Reversal Portfolio

A simple long–short reversal portfolio was constructed by

$$\text{Long} = \text{Lowest Momentum Stocks}$$

and

$$\text{Short} = \text{Highest Momentum Stocks}.$$

The portfolio return was therefore

$$R_{\text{LS}} = R_{\text{Low}} - R_{\text{High}}.$$

The resulting statistics were:

Table 4: Long–Short Reversal Portfolio

Metric	**Value**
Mean daily return	0.011876
Median daily return	0.008959
Daily standard deviation	0.055481
Annualized return	13.3780%
Annualized volatility	21.7113%
Sharpe ratio	0.6162

Inference

The long–short construction demonstrates that the negative momentum relationship can translate into a positive low-minus-high portfolio spread.

The resulting Sharpe ratio was approximately

$$\boxed{0.6162}.$$

However, these results should not be interpreted as a realistic live-trading performance estimate because transaction costs, market impact, financing costs, turnover, and other implementation effects were not included in this simple test.

12 Statistical Test of the Long–Short Spread

The mean low-minus-high spread was first tested using an ordinary t-test.

The result was

$$t = 8.1340$$

with

$$p = 8.86 \times 10^{-16}.$$

This appears to provide extremely strong evidence against a zero mean spread under the assumption of independent observations.

However, the forward returns used in the analysis overlap.

13 Overlapping-Return Problem

The 20-day forward returns overlap heavily.

For example, the return

$$R_{t,t+20}$$

shares most of its observations with

$$R_{t+1,t+21}.$$

Consequently, consecutive observations are not independent.

This means that an ordinary t-test can overstate the amount of independent information in the sample.

Inference

The ordinary p-value

$$8.86 \times 10^{-16}$$

should therefore not be interpreted without accounting for serial correlation caused by overlapping observations.

A heteroskedasticity and autocorrelation consistent correction is required.

14 HAC / Newey–West Adjustment

A HAC (Heteroskedasticity and Autocorrelation Consistent) / Newey–West adjustment was applied using 20 lags.

The adjusted results were:

Table 5: HAC / Newey–West Results

Statistic	**Value**
HAC standard error	0.003999
HAC t-statistic	2.9698
HAC p-value	0.002980

The statistical significance becomes weaker than under the ordinary t-test, but the result remains statistically distinguishable from zero under the HAC adjustment.

Inference

After accounting for the dependence created by overlapping 20-day returns, the reversal spread remains statistically significant:

$$\boxed{p \approx 0.00298}.$$

This strengthens the evidence that the observed reversal effect is not solely an artifact of the ordinary independent-observation t-test.

15 Evidence Across Different Periods

Additional observations of the relevant quantile relationship included IC values such as

$$-0.0519, \qquad -0.0642, \qquad -0.0926.$$

All of these values are negative.

Inference

The negative relationship was not confined to a single isolated period.

The direction of the relationship remained negative across the examined periods, providing additional evidence consistent with reversal.

16 Problems and Limitations

Several issues were identified during the analysis.

16.1 Missing Future-Return Column

During the horizon analysis, the code initially produced an error because the required future-return column had not yet been created.

The issue was resolved by explicitly calculating future returns using grouped forward shifts.

Inference

This was a coding/setup issue rather than an empirical finding. Once corrected, the horizon analysis could be performed consistently.

16.2 Extreme Daily Returns

Several stocks exhibited unusually large daily price movements.

These observations can influence

- return distributions,
- correlations,
- portfolio statistics,
- statistical tests.

They were retained rather than arbitrarily removed.

Inference

The results should be interpreted with awareness that extreme observations may contribute to the observed volatility and statistical relationships.

16.3 Overlapping Forward Returns

The 20-day forward returns overlap substantially, creating serial dependence between consecutive observations.

This makes the ordinary t-test inappropriate as the sole statistical test.

Inference

The HAC/Newey–West correction was therefore necessary.

The reversal spread remained statistically significant after this correction.

16.4 Multiple Lookback/Horizon Combinations

A total of

$$5 \times 4 = 20$$

lookback–forward-horizon combinations were tested.

The strongest mean IC occurred for

$$\text{20-day lookback} \rightarrow \text{20-day forward horizon.}$$

The corresponding IC was

$$-0.0673.$$

Because multiple combinations were tested, the strongest result should not automatically be regarded as an independently discovered effect.

Inference

The 20-day/20-day result is particularly interesting, but ideally requires validation on an independent out-of-sample period before being considered robust.

17 Momentum Versus Reversal

The original hypothesis was conventional momentum:

$$\text{Past Winners} \longrightarrow \text{Future Winners.}$$

This hypothesis implies

$$\text{IC} > 0.$$

However, the observed relationship was generally

$$\text{IC} < 0.$$

The empirical pattern was instead

$$\text{Past Winners} \longrightarrow \text{Lower Future Returns,}$$

while

$$\text{Past Losers} \longrightarrow \text{Higher Future Returns.}$$

This pattern is consistent with a **reversal effect** rather than conventional momentum.

18 Summary of Results

Table 6: Summary of Momentum Analysis

Test	**Result**	**Interpretation**
20-day momentum → 1-day return	IC = −0.01965	Negative relationship
5–120 day lookbacks	All mean ICs negative	No conventional momentum detected
20-day → 20-day	IC = −0.0673	Stronger reversal relationship
Quantile portfolios	2.1763% → 0.9887%	Future return decreases with momentum
Low-minus-high spread	1.1876%	Positive reversal spread
Ordinary t-test	$t = 8.13$	Strong apparent significance
HAC test	$t = 2.97$, $p = 0.00298$	Significant after overlap correction
Long–short Sharpe ratio	0.6162	Positive risk-adjusted spread in the simple test

19 Final Conclusion

The momentum analysis of `Dataset_PS-A.csv` does not provide evidence supporting conventional momentum over the tested horizons.

Instead, the results consistently show a **negative relationship between past returns and subsequent returns**.

The major findings are summarized below.

1. The initial 20-day momentum test produced a negative mean IC:

$$\boxed{\text{IC} = -0.01965}.$$

2. All tested combinations of 5, 10, 20, 60, and 120-day lookbacks with 1, 5, 10, and 20-day forward horizons produced negative mean ICs.

3. The strongest observed relationship occurred for

$$\boxed{\text{20-day lookback} \rightarrow \text{20-day forward}}$$

with

$$\boxed{\text{IC} = -0.0673}.$$

4. The quantile analysis showed a monotonic decline in future returns as past momentum increased:

$$2.1763\% \rightarrow 0.9887\%.$$

5. The resulting low-minus-high reversal spread was

$$\boxed{1.1876\%}.$$

6. The ordinary t-test produced

$$t = 8.134, \qquad p \approx 8.86 \times 10^{-16}.$$

   However, this test does not account for the dependence caused by overlapping 20-day returns.

7. After applying the HAC/Newey–West correction,

$$t = 2.9698, \qquad p \approx 0.00298.$$

   The relationship therefore remained statistically significant after accounting for the overlapping-return problem.

Overall, the evidence from this dataset is more consistent with

$$\boxed{\textbf{Short/Medium-Term Reversal}}$$

rather than conventional momentum.

In practical terms, within the tested sample and horizons, stocks that had performed relatively strongly in the recent past tended to subsequently underperform stocks that had performed relatively poorly.

However, this should not yet be interpreted as proof of a deployable trading strategy. Further validation would be required, including

- out-of-sample testing,
- transaction-cost analysis,
- turnover analysis,
- market-impact analysis,
- robustness checks across different market periods,
- and potentially correction for multiple hypothesis testing.

Final Research Statement

The momentum hypothesis was not supported by the tested OL-HCV data. Instead, the data exhibited a statistically significant negative relationship between past momentum and subsequent returns, with the strongest evidence around the 20-day lookback and 20-day forward horizon. Portfolio sorting confirmed a monotonic reversal pattern, and the low-minus-high reversal spread remained statistically significant after a HAC/Newey–West correction for overlapping returns. Thus, the principal finding of this analysis is evidence

of short/medium-term reversal rather than conventional momentum.